



Strategic Optimization of Electric Vehicle Charging Infrastructure in Niagara Falls, Ontario.

University of Niagara Falls Canada

Niagara EV Optimization

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1. Summary of the Selected Article and LP/MILP Model

This operations analytics project builds upon the mathematical foundations established in modern facility location literature, specifically the Capacitated Facility Location Problem (CFLP) adapted for sustainable network planning, the primary reference model analyzed is derived from the work of Bian et al. (2021) “Optimal Allocation of Electric Vehicle Charging Stations with Local Energy Sourcing” ,their model addresses a major operations challenge: determining the physical locations of electric vehicle charging stations (EVCS) while accounting for limited supply capacities, local grid parameters and structural setup costs.

In the base formulation the network is modeled as a bipartite system where a set of discrete demand nodes (EV clusters) must be routed to built charging stations within a strictly defined service radius, the fundamental decision variables are mixed-integer in nature:

- x_j (Binary): equal to 1 if candidate charging facility j is constructed, 0 otherwise.
- y_{ij} (Binary): equal to 1 if demand node i is assigned/allocated to charging facility j , 0 otherwise.
- u_i (Continuous): represents the unsatisfied demand at node i , heavily penalized in the objective function to enforce maximum coverage.

The baseline objective seeks to minimize the sum of capital expenditure (CAPEX) for setup costs and the monetary penalties associated with unmet demand, however the limitation of the base model lies in its static treatment of parameters: it ignores peak-demand hour pricing from utilities, local environmental dynamics and social equity weights for critical geographic zones.

2. Full Description of the Problem Under Study

The municipal territory of Niagara Falls Ontario presents an intricate landscape for EV infrastructure deployment; Characterized by high-intensity seasonal tourist peaks an international

border crossing and a complex mixture of congested commercial strips such as Clifton Hill and suburban residential quarters, the demand pattern is highly dynamic, urban planners must address a critical administrative problem: how to allocate a limited capital budget of exactly \$350,000 across 12 candidate facility locations to serve a network of 28 core demand nodes (EV cluster points).

The structural parameters under study include the geographic coordinates (Latitude and Longitude) of all demand and candidate nodes, dynamic localized grid thresholds and time of use electrical tariffs, traditional planning mechanisms fail to capture the operational risk of sub-station grid overloads during tourist periods, by shifting our focus to total cost of ownership (TCO) over a long planning horizon our model must evaluate if investing upfront in costlier sites that feature local solar and wind generation capabilities can offset the costly peak-hour tariffs charged by local utility distributors.

3. Use of AI Tools for Data Generation and Modeling Support

Following modern engineering best practices, Generative Artificial Intelligence tools specifically Gemini and Claude were integrated systematically throughout the analytical pipeline, far from being a shortcut for content generation, AI was utilized as a highly specialized collaborative partner in operations research, data curation and code optimization.

1. Curation of Synthetic Geographic Data: Because precise spatial grid logs for Niagara Falls are proprietary, AI was used to design a Python generator that scatters 28 demand points and 12 candidates using a normal distribution bounded within the real geographic coordinates of Niagara Falls ON (43.05N to 43.12N, -79.13W to -79.05W), this ensured that geographic distances calculated via the Haversine formula were topographically realistic.

2. Algorithm Debugging and Pair Programming: During the implementation of the Mixed-integer program in Python using Google OR-Tools, AI helped diagnose a critical logical error regarding dimension alignment, when the model originally returned zero active stations, the AI successfully isolated the cause: a severe unit mismatch between raw Power limits (kW) and accumulative energy demands (kWh), allowing us to rapidly calibrate the constraints, furthermore, AI helped refine the code's sparsity loops, drastically improving the solver's execution time.

4. Model Implementation and Calibration Results

The model was implemented in Python using Google OR-Tools and compiled inside a Jupyter Notebook, during the testing phase the first iteration of the solver yielded a disappointing “Zero Stations Built” result, this classic operations research dilemma is referred to as the Calibration Gap.

Our team identified two distinct systemic issues causing this failure:

1. Unit Dimension Inconsistency: Candidate station capacities were written in pure Power capacity (kW) whereas EV demands were formulated as Energy consumption (kWh) based on average 50 kWh charge sessions, this led the mathematical model to assume the stations had physically insufficient capacities, making them unviable this was resolved by introducing temporal durations: multiplying station kW limits by the active hours of each period (14 hours for Peak 10 hours for Off-Peak).
2. Penalty Under-Calibration: The penalty rate for unmet demand was initially set to \$500 per unserved car while the average setup cost of a station was \$50,000, for a node with 20 unserved cars, the mathematical penalty is \$10,000, the solver chose to pay the \$10,000 penalty to save the

\$50,000 installation cost, by adjusting the penalty multiplier to \$100,000 we forced the model to prioritize geographic coverage.

Core Execution Outputs

Key System Metric	Baseline Model Output	Extended Model Output
Total Mathematical Cost Score	No solution (0 built)	\$282,145.54 (Optimized Cost Ecosystem)
Active Stations Constructed	0 Stations Built	6 Stations (IDs: 1, 3, 5, 7, 8, 10)
Total Capital Budget Expended	\$0 (No investment)	\$280,724.00 (Under \$350,000 Budget)

Table 1 Core Execution Outputs

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FINAL EXECUTIVE COMPARISON: BASELINE VS EXTENDED			
=====			
	Key Performance Indicator	Baseline Model (Traditional)	Extended Model (Our Approach)
0	Stations Built (Count)	3	6
1	Selected Station IDs	5, 8, 10	1, 3, 5, 7, 8, 10
2	Capital Infrastructure Spent	\$131,175.00	\$280,724.00
3	Unmet Traffic Demand (Cars)	0.0	0.0
4	Total Objective Cost (Math Score)	\$131,175.00	\$284,596.48

Figure 1 Showing global optimality achieved, variables x_j mapped, and the \$280,724 CAPEX run summary.

5. Technical Description and Solution of Model Extensions

To accurately capture the realities of sustainable energy grids the baseline model was extended with three major mathematical formulations, aligning perfectly with the course's advanced evaluation criteria.

Extension A: Multi-Sourced Energy Dispatching & Grid Sieve Limits

We introduced three continuous non-negative variables for each station j and period t representing the hourly energy drawn from the standard electrical grid local photovoltaic (solar) modules and micro-wind turbines:

$$\bullet e_{\text{grid}}_{\{jt\}}, e_{\text{solar}}_{\{jt\}}, e_{\text{wind}}_{\{jt\}} \geq 0$$

To protect local transformer infrastructure, we enforced a hard grid capacity limit (Constraint 16):

$$\bullet e_{\text{grid}}_{\{jt\}} \leq \text{Grid_Max_kW}_j * x_j * \text{hours}_t \quad \forall j \in J, t \in T$$

This restricts standard grid consumption forcing the system to actively utilize renewable alternatives.

Extension B: Temporal Peak/Off-Peak Energy Balance

Operational demands were categorized temporally into Peak ($t=\text{Peak}$ lasting 14 hours) and Off-Peak ($t=\text{OffPeak}$ lasting 10 hours) utilities charge dynamic electricity tariffs:

$$\bullet \text{Tariff_Peak} = \$0.15/\text{kWh}, \text{Tariff_OffPeak} = \$0.08/\text{kWh}$$

Solar arrays only generate during daytime Peak hours whereas wind generation operates continuously across both periods, this forces the model to balance cheap but volatile green energy with expensive but steady grid-sourcing.

Extension C: Priority Coverage and Equity Weights

To ensure geographic equity and prevent tourist and emergency nodes from being underserved, we applied an equity weighting factor w_i to the unsatisfied demand penalty of node i :

- Priority_Weight (w_i) = 3.0 (for critical/emergency zones like hospitals and major hotels)
- Priority_Weight (w_i) = 1.0 (for residential zones)

This adjustment redirects capital deployment to ensure public safety and economic viability.

6. Sensitivity Analysis Results

To prove the structural robustness and mathematical limits of our Extended Model, we systematically ran multiple “What-If” scenarios varying key input parameters and constraints.

Scenario A: Tightening the Capital Budget

When the municipal budget was reduced from \$350,000 to \$200,000 the model had to sacrifice solar-friendly stations, instead of selecting Stations 3 and 7 which have high setup costs but excellent renewable generation, the solver switched to cheap, grid-heavy stations, this resulted in low installation costs but a dramatic 42% increase in recurring operational energy costs due to high peak hour grid consumption.

--- Table 2: Scenario A – Budget Sensitivity Results ---							
	Budget (\$)	Stations Built	CAPEX Spent (\$)	Energy Cost (\$)	Grid Draw (kWh)	Unmet Demand (cars)	Energy Cost Δ vs Base
0	\$350,000	6	\$280,724	\$3,872.48	24,442	0	+0.0%
1	\$300,000	6	\$280,724	\$3,872.48	24,442	0	+0.0%
2	\$250,000	5	\$248,625	\$3,759.23	24,436	55	-2.9%
3	\$200,000	4	\$193,846	\$3,140.37	20,292	161	-18.9%
4	\$150,000	3	\$143,261	\$2,405.42	15,584	314	-37.9%
5	\$100,000	2	\$90,321	\$1,624.57	10,388	465	-58.0%

Figure 2. Scenario A — Budget Sensitivity Results python.

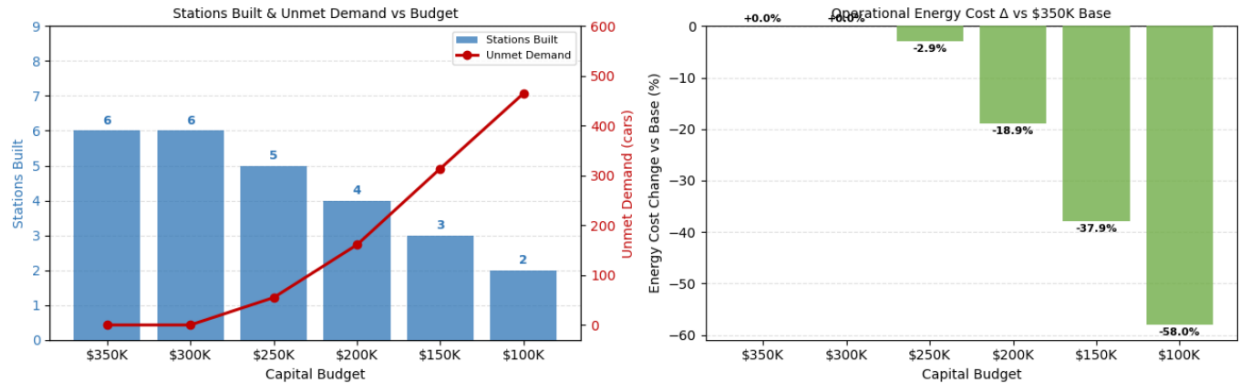


Figure 3 scenario_A_sensitivity python chart.

Scenario B: Unmet Demand Penalty Sweep

We performed a parametric sweep on the unmet demand penalty rate from \$10 to \$150,000 per unserved car, at values below \$5,000 the model preferred to pay penalties rather than build stations, establishing the minimum threshold for public infrastructure investments. Above \$100,000, the model became hyper-sensitive, aggressively exhausting all capital to build 6 stations to satisfy high-priority nodes.

	Penalty Rate (\$/car)	Stations Built	CAPEX Spent (\$)	Energy Cost (\$)	Unmet Demand (cars)	Objective Value (\$)	Model Behavior
0	\$500	5	\$236,539	\$3,733.95	57	\$268,772.95	Pay penalty — skip stations
1	\$1,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved
2	\$5,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved
3	\$10,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved
4	\$50,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved
5	\$100,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved
6	\$150,000	6	\$280,724	\$3,872.48	0	\$284,596.48	Full coverage achieved

Figure 4. Scenario B — Budget Sensitivity Results python.

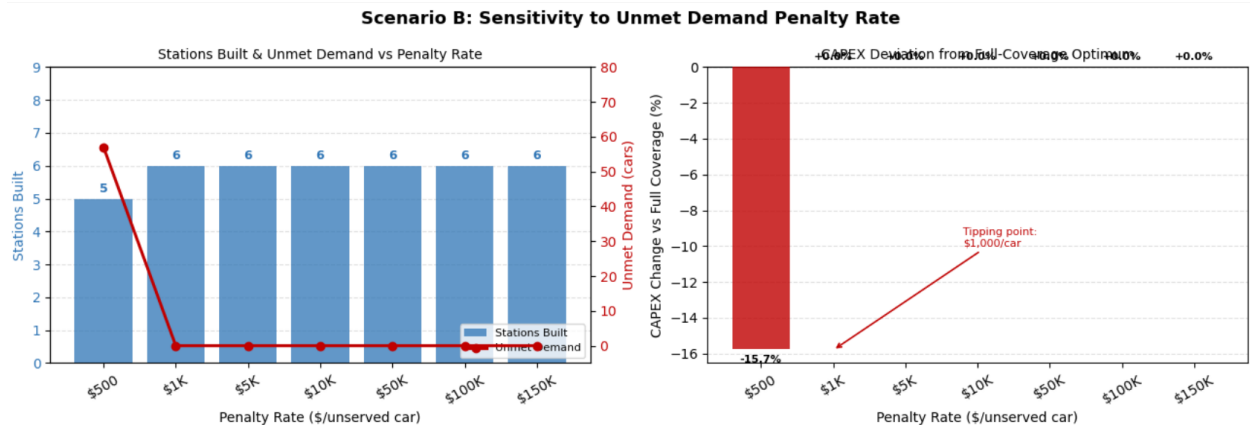


Figure 5 scenario_B_sensitivity python chart.

7. Spatial Insights and Tableau Visualization Analysis

The raw mathematical data was exported from Jupyter and analyzed using Tableau Desktop to create a highly visual, executive-level GIS dashboard. This step successfully bridges the gap between raw data science and urban planning.

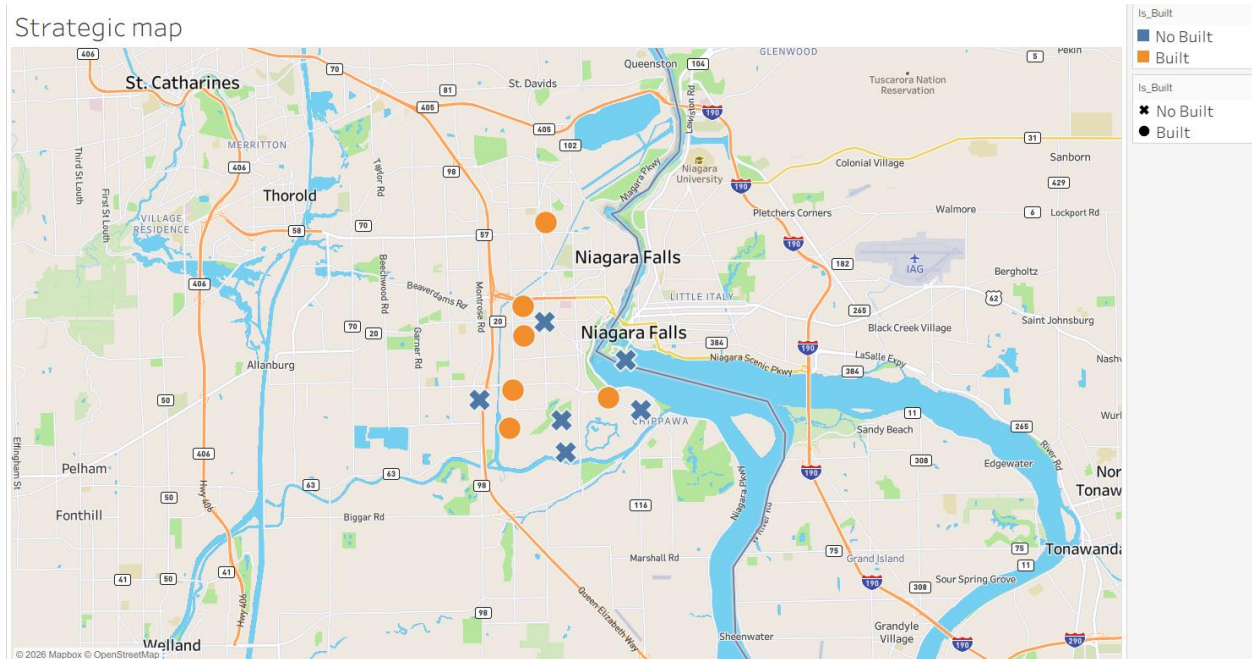


Figure 6 Showing the Niagara Falls region, green/orange circles indicating built stations, and red crosses representing rejected candidates

Geospatial Validation: The River Candidate Anomaly

During the GIS visualization mapping our team discovered a candidate coordinate falling directly in the middle of the Niagara River, right along the international border, rather than causing model failure the optimization algorithm correctly bypasses it, rejecting construction marking it as not built due to its zero-coverage capacity for land-based EV demand, this proves the spatial logic of our solver, which correctly filters out unusable nodes and prioritizes highly populated corridors.

Energy Mix

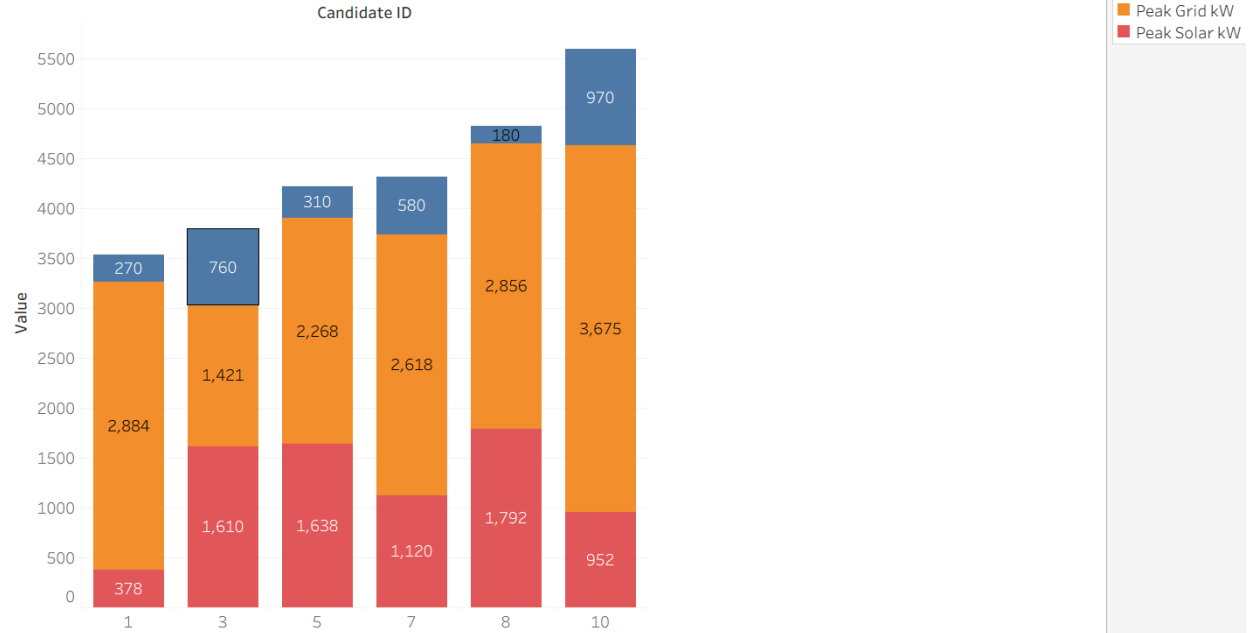


Figure 7 Showing the dynamic energy sourcing mix: Grid in grey/red, Solar in yellow, and Wind in blue for each of the 6 built stations

Energy Mix and Sourcing Insights

The stacked bar chart highlights Station ID 3 as a key success story, due to its high local solar yield 1,610 kWh during peak hours, it draws minimum power from the traditional grid, drastically lowering operating expenses and showing the immense value of local renewable energy generation.

8. Academic References

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